

## JAMB-Biology-Past-Questions-and-Answers

**Subject Area: JAMB Biology (UTME & Post-UTME)**

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### 1. EXAMINATION OVERVIEW & CORE OBJECTIVES

This document provides the essential study syllabus, high-frequency revision topics, and past examination questions compiled for candidates preparing for the Joint Admissions and Matriculation Board (JAMB UTME) and University Post-UTME screening tests. Comprehensive mastery of these concepts is recommended for achieving scores above 280+ in the unified examination.

### 2. HIGH-YIELD SYLLABUS TOPICS

- 1. Cell Structure and Functions (Organelles, Mitosis & Meiosis)
- 2. Mendelian Genetics, DNA Structure, Mutations & Heredity
- 3. Ecology: Ecosystems, Trophic Levels, Nutrient Cycles & Biomes
- 4. Physiology: Digestion, Circulatory System, Osmoregulation & Nervous Coordination
- 5. Plant Biology: Photosynthesis, Respiration, Transpiration & Auxins

#### CBT EXAM SUCCESS CHECKLIST:

1. Time Allocation: Aim for 45-50 seconds per question to leave 15 minutes for review.
2. Elimination Technique: Rule out two obviously incorrect distractors before final choice.
3. Biometric & Hall Verification: Always verify your computer seat number matches your slip.

### 3. CURATED PAST QUESTIONS & DETAILED SOLUTIONS

**1. Which organelle is primarily responsible for intracellular digestion and apoptosis?**

' **Option B: Lysosome**

*Explanation: Lysosomes contain hydrolytic enzymes that break down waste materials, cellular debris, and foreign particles.*

**2. In a Mendelian monohybrid cross between two heterozygous tall pea plants (Tt), what is the expected phenotypic ratio of tall to dwarf offspring?**

' **Option C: 3:1**

*Explanation: The genotypic outcome is 1 TT : 2 Tt : 1 tt, yielding 3 tall plants to 1 dwarf plant.*

**3. The functional unit of the mammalian kidney responsible for ultrafiltration is the:**

' **Option A: Nephron (specifically the Bowman's Capsule / Glomerulus)**

*Explanation: High hydrostatic pressure in the glomerulus forces water, ions, and small solutes across the podocyte membrane into Bowman's capsule.*

**4. Which of the following ecological succession processes begins on bare, previously uninhabited rock or volcanic lava?**

' **Option D: Primary Succession**

*Explanation: Primary succession starts in areas devoid of soil, typically initiated by pioneer species like lichens and mosses.*

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